

Evoked Potential and Visual Half-Field Evidence for Task-Dependent Cerebral Asymmetries in Congenitally Deaf Adults

VINCENT J. SAMAR

National Technical Institute for the Deaf at the Rochester Institute of Technology

Twelve congenitally deaf subjects participated in a visual half-field experiment under two conditions, lexical decision and line orientation discrimination. Reaction time, accuracy, and evoked potentials from left and right temporo-parietal sites were simultaneously obtained. The evoked potential data were analyzed using a principal components analysis followed by analysis of variance and multiple regression. Both the visual half-field and the evoked potential data produced evidence of task-dependent cerebral asymmetries. Individual differences in visual half-field asymmetries in accuracy and individual scores on three cognitive paper and pencil tests were strongly predicted by linear combinations of several independent hemispheric asymmetry variables derived from evoked potential component scores. The results suggest that cerebral specialization in congenitally deaf people is a complex, multidimensional phenomenon, and provide the first evidence for a direct relationship between cerebral asymmetry and cognitive performance in this population.

Studies of the relationship between congenital deafness and cerebral organization are of interest to researchers and professionals in several diverse disciplines. Researchers interested in the interplay of biological predisposition and experiential factors have found that the study of cerebral organization in deaf subjects can provide valuable insights concerning this issue. For example, it has recently been reported that congenitally deaf and hearing subjects show substantially different patterns of neuroelectric activity evoked by visually presented words (Neville, Kutas, & Schmidt, 1982b) as well as by simple, nonlanguage visual stimuli (Neville, Schmidt and Kutas, cited in Neville et al., 1982b). These data have been taken to suggest that sensory deprivation may alter the normal

This research was conducted at the National Technical Institute for the Deaf at the Rochester Institute of Technology in the course of an agreement with the U.S. Department of Education. Send requests for reprints to Vincent J. Samar, Department of Communication Research, The National Technical Institute for the Deaf, Rochester, NY 14623.

course of development of cerebral mechanisms related to perceptual and cognitive functions. To clinicians interested in the remediation and training of the communication skills of deaf children, the study of cerebral organization holds the promise of understanding the role of sensory deprivation and language experience in influencing the neurological substrates of communication skills. Such information may have implications for the educational practices employed with deaf students, including the structuring of communication training programs and assessment techniques. Similarly, educators concerned with the academic development of deaf children have an interest in understanding the role of environmental and experiential factors in determining the cerebral organization of deaf children and in the implications for the development of cognitive style and performance in the academic and professional setting.

Currently, our understanding of the effects of congenital deafness on cerebral organization is at a relatively primitive stage. Recently, some attempts have been made to study one important aspect of cerebral organization in the deaf population, hemispheric specialization (Kelly & Tomlinson-Keasey, 1977; Manning, Goble, Markman, & LaBreche, 1977; McKeever, Hoemann, Florian, & Van Deventer, 1976; Neville & Bellugi, 1978; Phippard, 1977; Poizner, Battison, & Lane, 1979). These studies have generally relied upon the visual half-field technique and have compared hearing and deaf subjects. These studies have variously reported similar, equivalent, or opposite visual half-field asymmetry patterns between deaf and hearing subjects on a variety of English, sign language, and spatial tasks. Unfortunately, the heterogeneity of subject characteristics and results in this body of research, the lack of precise replication conditions, and the presence of numerous methodological problems tend to preclude any convincing conclusion with respect to the presence or nature of functional cerebral asymmetries in the deaf population. (See Poizner & Lane, 1979, for a review.)

The lack of conclusive evidence on this issue may be due, in part, to limitations of the techniques previously employed to study cerebral specialization in the congenitally deaf population. While the visual half-field technique is an important contemporary method of investigating cerebral specialization, it is likely that this technique is not fully adequate to capture the complexity of the cerebral asymmetry phenomenon. A number of recent electrophysiological studies have demonstrated that information processing in neurologically normal, hearing subjects is associated with the activity of numerous independent cerebral mechanisms during a given task (eg. Molfese, 1978, 1979; Molfese & Hess, 1978). Samar (1983) presented electrophysiological evidence that visual half-field asymmetries may be multiply influenced by several underlying cerebral mechanisms which display independent lateralization distributions across the population. This implies, for example, that equivalent visual field performance between

deaf and hearing groups may not necessarily reflect equivalent underlying patterns of hemispheric involvement. Furthermore, the absence of field advantages on some tasks in deaf groups may not necessarily reflect bilaterally equivalent hemispheric processing activities. Rather, such results may be the product of the operation of several distinctly but differently lateralized processing events which tend to have a canceling influence on the resultant visual half-field asymmetries. These possibilities make conclusions about the influence of deafness on cerebral specialization based on visual half-field data very uncertain, especially in light of the inconsistency of results across previous studies.

The present study is concerned with cerebral specialization *within* a congenitally deaf sample. Similarities or differences between deaf and hearing subjects are not a concern in this research. Certain considerations limit the inferential power of such comparisons at the present time. Deaf and hearing subjects differ markedly on a number of dimensions other than hearing status. Such variables include the type and extent of exposure to language systems, personality variables such as field dependence-independence, perceptual processing strategies, etc. (Parasnis & Long, 1979; Parasnis and Samar, 1982), all of which might affect the organization and/or manifestation of functional cerebral asymmetries. For example, some recent evidence suggests that bilinguals display different patterns of cerebral asymmetry than monolinguals (Walter and Zatorre, 1978). Deaf children, particularly those subjects chosen for previous laterality studies, often are exposed to two or more languages as they develop, including ASL or Pidgin Signed English and some visual form of direct English such as print, finger spelling, or speech presented through lip and facial movements. However, previous studies comparing deaf and hearing individuals have generally employed monolinguals as the hearing comparison group, thereby confounding polylingualism with deafness. This confound is further complicated by the fact that severely and congenitally deaf students have primarily acquired their languages through the visual modality and typically do not achieve proficiency in languages which originally evolved through the auditory modality such as English (Trybus & Karchmer, 1977; Moores, 1978). Such considerations make the identification of the specific effect of auditory deprivation on cerebral organization, through the comparison of deaf and hearing adults, problematic at best.

The present research was undertaken for two specific reasons: (a) to examine the correspondence between an independent measure of cerebral asymmetry (evoked potential data) and visual half-field asymmetry in congenitally deaf subjects, and (b) to explore the possibility that individual differences in cerebral asymmetry are related to individual differences in performance on certain independent tests of cognitive skills.

METHODS

Subjects

Sixteen right-handed congenitally and severely deaf subjects from the National Technical Institute for the Deaf (NTID) were tested. Eight were males and eight were females. However, four subjects were dropped prior to analysis due to the presence of eye movement artifact in the EEG records. The 12 subjects used lost their hearing prelingually, either at birth or prior to 2 years of age. The mean pure tone hearing loss in the better ear (tested at 500, 1000, and 2000 Hz) was 96.0 dB with a standard deviation of 7.9 dB and a range of 80 to 107 dB.

Ten of the twelve subjects had no deaf relatives. The remaining two had deaf parents. All subjects learned to sign prior to 5 years of age, as assessed by a language background questionnaire (Hatfield, Caccamise, & Siple, 1978) in routine use at NTID. All subjects were proficient at manual communication as assessed by the CID Manual Communication Test (Johnson, 1975). The mean Manual Communication Test score was 86.5% with a standard deviation of 6.9% and a range of 78 to 98%.

Handedness was assessed by means of a questionnaire which contained 15 questions similar to those from the Edinburg Inventory for Handedness, such as "Which hand do you write with?" and "Which hand do you use to hold a hammer when hammering a nail?". The subject responded to all 15 questions by circling 1 of the following choices: left, right, both, don't know. The mean number of right hand responses to the 15 questions was 12.73 with a standard deviation of 1.34, indicating a strong right hand preference in all subjects. All subjects wrote and drew with their right hand.

Tasks

The experiment included a Linguistic and a Spatial Condition. The Linguistic Condition was a two-choice lexical decision task in which subjects discriminated between three-letter words and nonsense syllables. The Spatial Condition was a two-choice line orientation discrimination task in which subjects determined whether two simultaneously presented lines were of the Same or a Different orientation. Several authors have previously reported right and left visual field performance advantages, respectively, on these two categories of discrimination tasks for hearing subjects (e.g., Day, 1978; Hines, 1976; Bradshaw & Gates, 1978; Fontenot & Benton, 1972; Sasanuma & Kobayashi, 1978).

Stimuli

The linguistic stimuli were (a) 24, high-frequency, three-letter words chosen from the norms of Kucera and Francis (1967) such that the frequency of occurrence was 360 or more, and (b) 24 pronounceable nonsense syllables. The nonsense syllables were derived from the words by substituting one letter such that the consonant-vowel class of the substituted letter was preserved. The specific stimuli used have been presented previously (Samar, 1983). The spatial stimuli were 24 pairs of lines having identical orientation (same pairs), ranging from 0° to 345° in 15° steps, and 24 pairs of lines, distributed over the same orientation range, and differing in orientation from each other by 15° (different pairs). All stimuli were drawn on 4 × 6-in. cards in a vertical orientation, and presented 3.5° from fixation. Words were printed in capital letters, and were 1.26° high × 0.32° wide. Lines were presented one above the other, separated by 4° at their center, and were 1.26° long.

Each linguistic stimulus was reproduced four times, twice in the right visual field and twice in the left visual field, for a total of 192 stimulus cards. Each of the spatial stimuli was reproduced four times, twice in each field. Because of the symmetry constraints inherent in the construction of the spatial stimuli each stimulus in each field was matched with an identical stimulus in the other field and a mirror image stimulus. Detailed explanation of stimulus construction is given elsewhere (Samar, 1983).

Design

The experimental design involved (2) Sexes, (2) Conditions, (2) Halves of each condition, (2) Fields of stimulation, (2) Stimulus Types in each condition (Word vs. Nonsense Syllable and Same vs. Different), and (2) Hemispheres from which evoked potentials were recorded. This resulted in the acquisition, from each subject, of 16 average reaction time scores for correct responses and 16 corresponding accuracy scores. Similarly 32 averaged evoked potentials for correct responses were acquired from each subject, 16 from each hemisphere corresponding to the various treatment combinations.

The 192 stimuli per condition were arranged in eight blocks of 24 trials each, in each condition. Extensive randomization and counterbalance techniques were employed to insure the nonpredictability of Field of Presentation and Stimulus Type from trial to trial, and to equate the structure of sequential dependencies in each visual half-field (see Samar, 1983).

Procedure

The experiment was controlled by a Nicolet Med 80 laboratory computer and the stimuli were presented on a four-field Gerbrands tachistoscope. The computer initiated a trial every 6 sec. Each trial consisted of a 1-sec foreperiod during which a fixation point appeared, followed by a stimulus 50 msec in duration. A blank white field was present between trials. All fields were calibrated for an equal luminance value of 3.50 ft-L.

Subjects were screened to insure that they had normal or corrected vision and had no history of neurological disorder apart from sensory-neural deafness. Silver cup electrodes were placed at left and right temporal-parietal locations, 30% of the intermeatal distance down from vertex and 15% of the nasion-inion distance back from the intermeatal line. These two electrodes were referenced to linked earlobe leads. A ground electrode was placed on the forehead, and EOG electrodes were placed at the outer canthus of the left eye and near the bridge of the nose. All electrode impedances were maintained at under 2 kohm.

Prior to each subject's session, a 50- μ V calibration signal was fed through the entire system, beginning at the input to the EEG amplifiers. The computer acquired and digitized these calibration signals and stored them on floppy disk for later use in converting the averaged evoked potential waveforms to the microvolt metric. Since the computer digitized the calibration signals from a common source separately for each of the two hemisphere channels, differences in absolute amplification due to individual channel differences were taken into account in converting the raw evoked potential data to microvolt values.

Subjects were seated in a comfortable chair and tested in a sound attenuating, electrically shielded IAC chamber. Each subject participated in both conditions in a single session. Half of the subjects received the Linguistic Condition first and half received the Spatial Condition first. Within these two groups, half of the subjects used the index finger of their right hand to respond to Words and Same orientations, and the middle finger to respond to Nonsense Syllables and Different orientations. The other half used the reverse order of fingers for these responses. Within each of these four groups, half of the subjects were male and half were female.

A 24 trial practice block was presented before each condition. During this time, it was confirmed that the subject understood the task, that eye movements were well controlled, and that the recordings were free of electrical artifact. During the experimental run, a 3-min break was given between conditions. A 1-min break was given between all other blocks.

The EEG and EOG were amplified by Grass 7P511 ac preamplifiers. The computer simultaneously presented the stimuli, acquired reaction time and accuracy data, and acquired and averaged the evoked potentials and EOG samples. The evoked potentials and EOGs were averaged into one of four bins corresponding to the four trial types (2 Stimulus Types $\times$ 2 Fields) in each condition. Only correct trials were accepted for the averages. These

averaged waveforms were digitally filtered by the computer with a flat bandpass of 0.1 to 30 Hz. Frequencies between 30 and 50 Hz were sharply attenuated by a cosine bell transfer function and all frequencies above 50 Hz were eliminated.

Cognitive Tests

Scores on three subtests of the Differential Aptitude Tests (Bennett, Seashore, & Wesman, 1966) were obtained for each student from student files. The three tests used were the Verbal Reasoning Test, the Spatial Relations Test, and the Abstract Reasoning Test. The Verbal Reasoning Test is a test of facility in solving 50 verbal analogies requiring vocabulary skills and the appreciation of logical and analogical relations among verbally expressed concepts. The subject solves an analogy by choosing one of four alternative verbal responses. The Spatial Relations Test requires subjects to visualize and mentally manipulate 60 two-dimensional foldups of three-dimensional nonnameable geometric objects, and match them to one of four alternative pictorial representations. The Abstract Reasoning Test requires subjects to discover logical progressive relationships among four pictorial presentations of nonnameable objects and to choose a fifth pictorial frame to complete the sequence from among five alternatives. The test is composed of 50 items. The Abstract Reasoning Test was developed as a test of nonverbal intelligence.

RESULTS

Four subjects were eliminated prior to data analysis due to the presence of eye movement artifact. One was male and three were female. Subsequent results are based on analyses of the remaining 12 subjects.

Behavioral Data

The group reaction time (RT) and accuracy means are presented in Table 1. A five-way analysis of variance involving (2) Sexes $\times$ (2) Conditions $\times$ (2) Presentation Halves $\times$ (2) Visual Fields $\times$ (2) Stimulus Types were performed for the RT and accuracy data separately. Sex was a between subjects treatment. Condition, Presentation Half, Visual Field, and Stimulus Type were repeated measures. Stimulus Type was nested under Condition and all other treatments were crossed. There were significant main effects for Condition ($F(1, 10) = 5.53, p < .05$) and Presentation Half ($F(1, 10) = 16.88, p < .005$) for the reaction time data. Subjects were overall 129.89-msec faster in the Spatial Condition than in the Linguistic Condition. Also, subjects were overall 95.05-msec faster in the second half of the conditions than in the first half. There was also a significant Condition $\times$ Half interaction ($F(1, 10) = 7.90, p < .02$). This interaction was due to a more dramatic drop over condition half in overall RT in the Linguistic Condition (160.37 msec) than in the Spatial Condition (29.71 msec). In each condition, there was a main effect for Stimulus Type in RT (Spatial Condition, $F(1, 10) = 8.75, p < .02$; Linguistic Condition, $F(1, 10) = 9.45, p < .02$). In the Spatial Condition, subjects responded to Same stimuli on the average 126.42-msec faster than to Different stimuli. In the Linguistic Condition subjects responded to word stimuli on the average 199.85-msec faster than to Nonsense Syllable stimuli. The Condition $\times$ Field interaction for the

TABLE I
GROUP MEAN REACTION TIME (RT) AND ACCURACY (% CORRECT) DATA FOR EACH CONDITION

	Spatial Condition				Linguistic Condition			
	Left Field		Right Field		Left Field		Right Field	
	Same	Different	Same	Different	Word	Nonword	Word	Nonword
First half								
RT	890.7	1040.1	906.5	1042.3	1020.8	1275.3	1070.1	1294.3
% Correct	94.1	67.7	87.8	66.0	83.7	79.9	80.2	85.4
Second half								
RT	881.7	948.5	888.5	1042.1	930.7	1088.1	918.4	1081.8
% Correct	93.1	69.4	91.3	62.8	85.8	80.2	85.8	87.5
Total								
RT	886.2	994.3	897.5	1042.2	975.8	1181.7	994.3	1188.1
% Correct	93.6	68.6	89.6	64.4	84.8	80.1	83.0	86.5

RT data was not significant, nor was there any significant overall main effect of Field. Hence, the RT data provide no evidence for task dependent visual field asymmetries in this subject sample.

For the accuracy data, there was a significant main effect for Condition ($F(1, 10) = 9.16, p < .02$). On the average, subjects correctly discriminated 79.04% of the spatial stimuli and 83.55% of the linguistic stimuli. In the Spatial Condition, there was a highly significant main effect for Stimulus Type ($F(1, 10) = 32.72, p < .001$). Subjects were considerably more accurate in responding to Same stimuli (91.58% correct) than to Different stimuli (66.49% correct). There was also a highly significant Condition $\times$ Field interaction ($F(1, 10) = 11.66, p < .01$). This interaction is presented in Table 2.

The table indicates a 4.1% left field advantage in accuracy in the Spatial Condition and a 2.3% right field advantage in accuracy in the Linguistic Condition. Simple effects analysis revealed that the Spatial Condition left field advantage was statistically reliable ($F(1, 10) = 5.15, p < .05$), while the Linguistic Condition right field advantage barely missed the conventional .05 level of significance ($F(1, 10) = 4.87, p = .052$). No other main effects or interactions were significant.

Evoked Potential Data

The evoked potential data were submitted to a Principal Components Analysis (Chapman, McCrary, Bragden, & Chapman, 1979). The EP data set consisted of 384 averaged EPs (12 subjects $\times$ 32 averaged EPs) which had been digitized at the rate of 125 Hz (8 msec per point) over 640 msec (80 time points) including a 48-msec pre-stimulus baseline period. Each EP was baseline corrected by subtracting the average amplitude value of the first six prestimulus data points from that of the remaining 74 points. The data were then arranged in a 384 (EP cases) $\times$ 74 (post-stimulus time point amplitude variables) matrix and submitted to the BMDP4M factor analysis program. The program derived a 74 $\times$ 74 product-moment correlation matrix and then extracted the principal com-

TABLE 2
OVERALL AVERAGE LEFT AND RIGHT VISUAL HALF-
FIELD ACCURACY SCORES (% CORRECT) IN
THE SPATIAL AND LINGUISTIC
CONDITIONS

Field	Condition	
	Spatial	Linguistic
Left	81.1	82.5
Right	77.0	84.8
Difference	4.1	-2.3

ponents. Unity was retained in the correlation matrix diagonal and only components with eigenvalues greater than one were retained and rotated using the normalized varimax rotation criterion. Factor scores for each EP on each component were provided by the program and were used in subsequent statistical analysis.

This analysis produced 11 components which collectively accounted for 89.6% of the total EP variance. These components, along with the grand average EP over all subjects and treatments (Centroid) are shown in Fig. 1. The component plots indicate the period of time during which each component maximally contributed to the EP waveform amplitude. The centroid was characterized by a positive peak at 112 msec (P112) followed by a negative peak at 152 msec (N152). A marked positive inflection occurred next at approximately 240 (P240) msec on the ascending

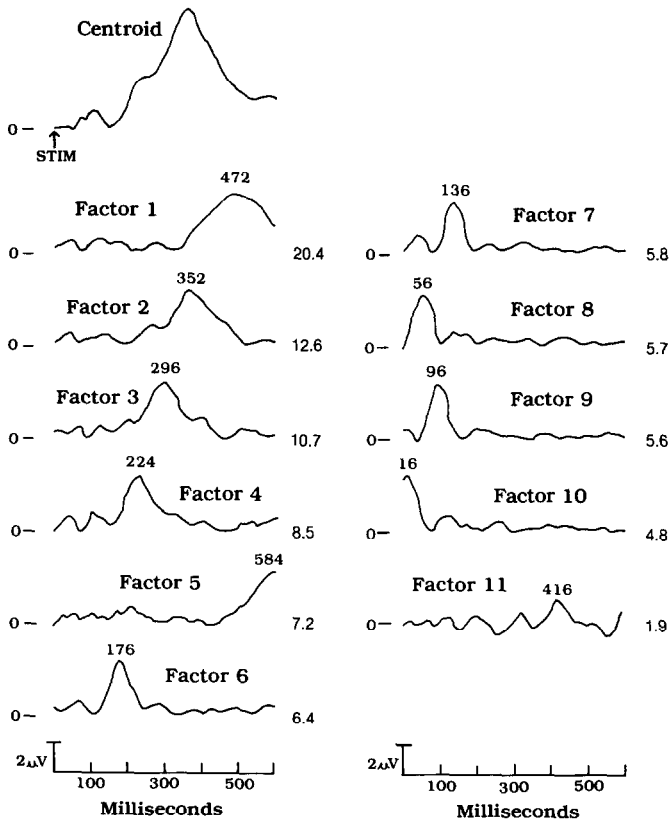


FIG. 1. Centroid (grand average EP over 384 cases) and 11 principal components. Positive is up. Factor plots represent the fundamental time courses of the factors (factor loadings multiplied by the standard deviations of the original variables). Numbers above plots are peak latencies in milliseconds. Numbers at the right refer to the percentage of the total EP variance accounted for by each factor.

slope of a prominent late positivity (P344). This peak was followed by a late negativity at approximately 560 msec (N560).

To examine the functional significance of the components, each set of factor scores for each component was submitted to a separate six-way analysis of variance. The analysis model for these ANOVAS was identical to that for the behavioral data presented above, with the exception of the addition of a repeated measures Hemisphere treatment.

Since the components are statistically independent, 11 independent ANOVAS are appropriate. However, such an analysis produces several hundred independent tests of partitioned variance, some of which may produce "significant" results due to extreme sampling error. To reduce the possibility of Type I errors it was decided to ignore interactions higher than the three-way term and to adopt a conservative alpha level of .01 for the ANOVA comparisons.

Factor 1 showed a main effect for Condition ($F(1, 10) = 13.33, p < .005$). The effect of this factor was to increase the negativity of N560 in the Linguistic Condition relative to the Spatial Condition. Factor 1 also showed a significant nested main effect for Stimulus Type in the Linguistic Condition ($F(1, 10) = 12.4, p < .01$). The effect of the component was to increase the negativity of N560 for Word stimuli relative to Nonsense Syllable stimuli. Factor 2 showed a main effect for Presentation Half ($F(1, 10) = 17.31, p < .005$). This component reduced the positivity of the prominent P344 peak in the second half of the both conditions relative to the first half of the conditions. Like Factor 1, factor 2 showed a nested Stimulus Type main effect ($F(1, 10) = 11.57, p < .01$). However, this effect was operative in the Spatial Condition, with the result that the positivity of P344 was reduced for Different stimuli compared with Same stimuli. Factor 2 also showed a significant Presentation Half $\times$ Stimulus Type interaction in the Spatial condition ($F(1, 10) = 13.64, p < .005$). This interaction is shown in Fig. 2. The interaction was due to a sharp

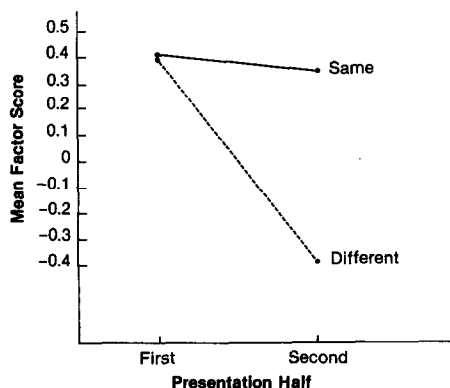


FIG. 2. Presentation Half $\times$ Stimulus Type interaction in the Spatial Condition shown by factor 2.

divergence of the factor scores to Same versus Different stimuli in the second half of the Spatial Condition, having the effect of markedly reducing the positivity of P344 for Different stimuli relative to Same stimuli as familiarity with the task increased.

Factor 3 which, along with factor 2, influenced the P344 component, showed a highly significant Presentation Half $\times$ Stimulus Type $\times$ Sex interaction in the Linguistic Condition ($F(1, 10) = 29.44, p < .001$). This interaction is shown in Fig. 3. The interaction was characterized by greater discrimination of Words and Nonsense Syllables by males during the first half of the condition relative to the second half, but greater discrimination of the two classes of linguistic stimuli by females in the second half of the condition relative to the first half. In general, the effect of the component was to increase the positivity of the ascending slope of P344 for Words relative to nonsense syllables during that half of the condition showing maximum discrimination by each sex. Factor 4 showed a main effect for Sex ($F(1, 10) = 15.215, p < .05$). This component had the effect of increasing the positivity of the inflection on the ascending slope of P344 for females relative to males.

Factor 5, occurring quite late (584 msec) displayed a Stimulus Type $\times$ Hemisphere $\times$ Presentation Half interaction in the Spatial Condition ($p(1, 10) = 10.62, p < .01$) shown in Fig. 4. During the first half of the Spatial Condition, right hemisphere component scores maximally distinguished between Same and Different line orientation stimuli. However, by the second half of the Spatial Condition left hemisphere factor 5 scores to Same and Different stimuli were markedly different, while right

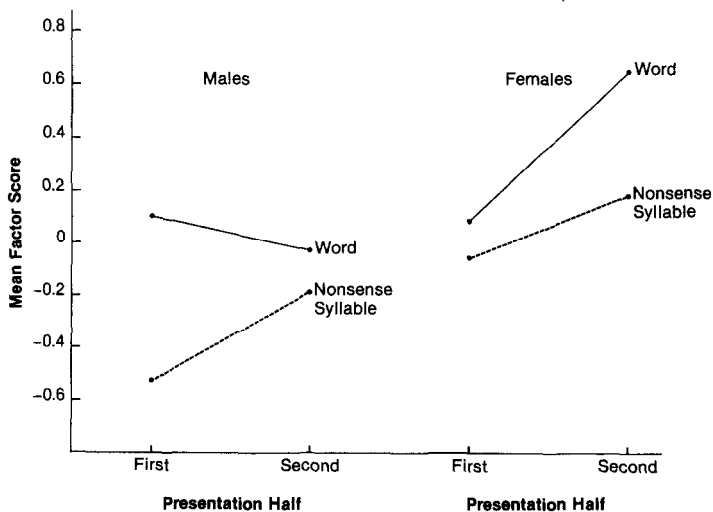


FIG. 3. Presentation Half $\times$ Stimulus Type $\times$ Sex interaction in the Linguistic Condition shown by factor 3.

hemisphere factor 5 scores were quite similar. The effect of factor 5 was to reduce the negativity of the N560 component for Different stimuli relative to Same stimuli in the hemisphere showing the greatest discrimination during each half of the condition. These data suggest a shift in the locus of hemispheric discrimination during the spatial task which most likely depends upon familiarity with the task and stimulus structure. Curiously, this hemispheric shift was not paralleled by any shift in field advantage in the behavioral data. The hemispheric discrimination pattern shown in Fig. 4 is not readily apparent in the surface features of the averaged EP waveforms. Figure 5 displays plots of the averaged waveforms during the first and second halves of the Spatial Condition for the left and right hemispheres separately with the EPs to Same and Different stimuli superimposed in each case. It is clear from inspection of the N560 component at the terminal region of the waveforms that the systematic change in differential hemispheric response to the two classes of stimuli revealed by factor 5 is masked by the presence of uncontrolled variance contributed by other PCA components. As such, it is unlikely that conventional peak amplitude measurement would suffice to reveal the processing event represented by factor 5. Factor 6 showed a *Field* $\times$ *Hemisphere* interaction ($F(1, 10) = 16.90, p < .005$) displayed in Fig. 5, which was significant at the .01 level or better in each of the conditions separately. The figure shows that the component scores to direct pathway stimulation (LF–RH and RF–LH) were more negative than those to indirect pathway stimulation (LF–LH and RF–RH). The effect of factor 6 was to increase the negativity of the N152 peak during direct pathway stimulation. This result is consistent with several previous studies which report a component

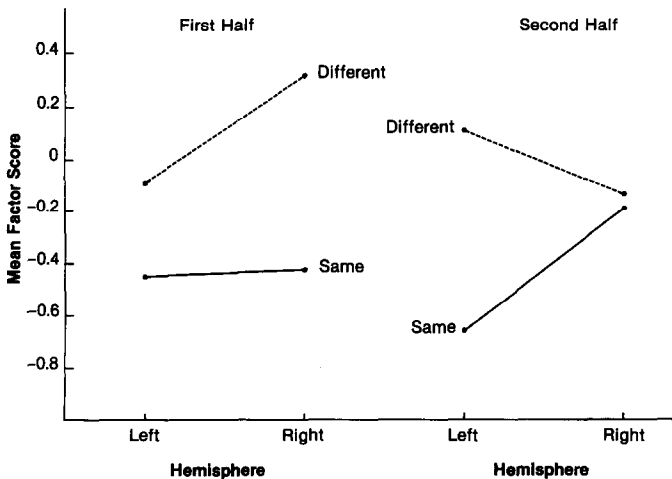


FIG. 4. Presentation Half $\times$ Stimulus Type $\times$ Hemisphere interaction in the Spatial Condition shown by factor 5.

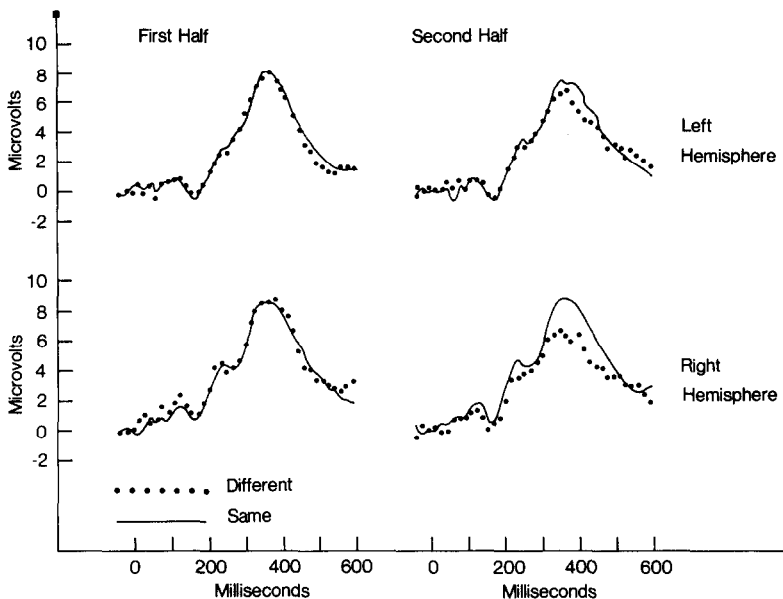


FIG. 5. Superimposed average EP waveforms for Same and Different stimuli from the first and second halves of the Spatial Condition for the left and right hemisphere leads.

in the time range of 150–200 msec which is sensitive to the anatomical properties of the visual system pathways (Ledlow, Swanson, & Kinsbourne, 1978; Neville, Kutas, & Schmidt, 1982a; Samar, 1983).

Factor 8 showed a highly significant *Presentation Half* $\times$ *Hemisphere* interaction ($F(1, 10) = 19.64, p < .001$). This interaction is shown in Fig. 6. The figure shows that the left and right hemisphere factor 8 scores reversed their relative polarity from the first to the second half of the conditions. Factor 8, which affects the immediate poststimulus portion of the evoked response appears to reflect a reversal of hemispheric activity as a function of familiarity with task demands which is independent of the nature of the task.

EP-Visual Field Performance Correlations

In a previous study on hearing subjects using the same experimental paradigm (Samar, 1983), multiple regression analyses revealed that visual field advantages in accuracy were strongly related to hemispheric asymmetry on a number of (independent) principal components. These data were interpreted to suggest that field advantages in the visual half-field paradigm may be multiply influenced by several underlying processing operations which have independent lateralization distributions across the population. To determine whether visual half-field asymmetries in deaf subjects are related to underlying hemispheric asymmetries, a similar

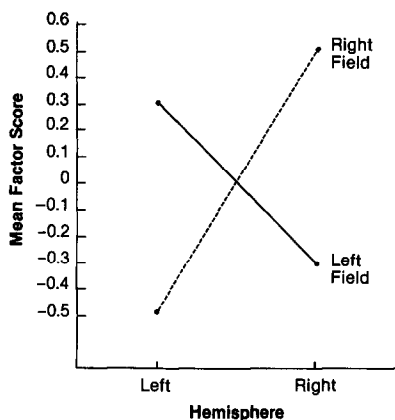


FIG. 6. Field $\times$ Hemisphere interaction shown by factor 6.

multiple regression analysis was conducted on the present data. Asymmetry scores were computed for each subject, for the visual half-field data and for the EP data, separately. For the visual half-field data, each subject received a field difference score for RT and for accuracy in the Linguistic and Spatial Conditions separately. This score was computed as the overall left field score minus the overall right field score in a particular condition, collapsing across Presentation Half and Stimulus Type. This resulted in four sets of visual half-field asymmetry scores (2 Conditions $\times$ 2 response measures) which were used as the dependent variables in four separate stepwise multiple regression analyses. For the EP data, a hemisphere difference score was computed for each subject for each condition and for each of the 11 principal components. This score was the overall left minus right hemisphere factor score for a particular principal component in a particular condition, collapsed over Presentation Half, Stimulus Type, and Visual Field. This resulted in two sets of 11 hemispheric asymmetry variables, one for each condition, which were employed as

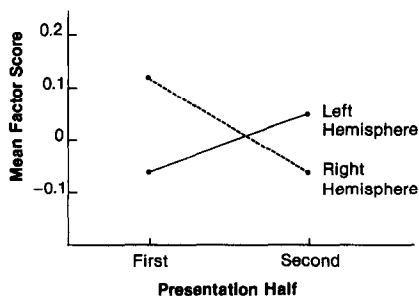


FIG. 7. Presentation Half $\times$ Hemisphere interaction shown by factor 8.

the independent variables in the regression analyses. The analyses were conducted *within* Condition. Thus, the 11 hemispheric asymmetry variables derived from the Spatial Condition were used to predict the RT or accuracy visual half-field asymmetry variables from that condition. Similarly the 11 hemispheric asymmetry variables derived from the Linguistic Condition were used to predict the visual half-field asymmetry variables from that condition. Because the EP data were collapsed across visual field (i.e., are independent of the spatial location of the stimulus) these analyses test the basic assumption of the visual half-field paradigm that overall visual half-field asymmetries reflect underlying hemispheric asymmetries in a given task.

Reaction time visual half-field asymmetry was not significantly predicted by any combination of hemispheric asymmetry variables in either condition. However, visual half-field differences in accuracy were strongly predicted by hemispheric asymmetry variables in each condition. Table 3 presents the results of the regression analyses. In the Spatial Condition, visual half-field asymmetry scores were significantly predicted by a combination of hemispheric asymmetry variables on factors 1, 6, 8, 2, and 11 ($F(5, 6) = 11.14, p < .01$). In the Linguistic Condition visual half-field asymmetry scores were significantly predicted by a combination of hemispheric asymmetry variables on factors 9 and 4 ($F(2, 9) = 7.09, p < .015$). These results are consistent with previous research on hearing subjects (Samar, 1983) which indicates that visual half-field accuracy asymmetry is multiply predicted by several independent hemispheric asymmetry variables in these two tasks.

TABLE 3
HEMISPHERE FACTOR SCORE DIFFERENCE VARIABLES ENTERING REGRESSION EQUATIONS TO
PREDICT VISUAL HALF-FIELD ACCURACY DIFFERENCE SCORES IN THE SPATIAL AND
LINGUISTIC CONDITIONS

Spatial Condition				Linguistic Condition			
PO	Lat	VE	Beta	PO	Lat	VE	Beta
1	472	31	-1.197	9	96	38	-0.549
6	176	19	-0.934	4	224	22	0.480
8	56	17	-0.593				$R = .78$
2	352	7	1.006				
11	416	16	0.926				
			$R = .95$				

Note. The predictor order (PO) columns list the numerical names of the electrophysiological factor from which the hemisphere difference scores were derived, in order of their entry into each equation. Lat, peak factor latency; VE, percentage variance explained; R , multiple correlation coefficient. Each beta weight is significant at the .05 level or better.

Relationships Between Asymmetry Measures and Cognitive Tests

A series of correlational analyses were run between the three DAT test scores and the behavioral and electrophysiological indices of cerebral asymmetry described above. Pearson product-moment coefficients were computed between each DAT test and the overall half-field difference scores in each condition. None of the six correlation coefficients (3 DAT tests $\times$ 2 Conditions) reached significance at the .05 level.

For the evoked response data, the 11 sets of hemispheric factor score difference variables from each condition were used to predict the DAT variables. No significant prediction was achieved for any DAT test when the hemispheric asymmetry variables from the Spatial Condition were employed as predictors. By contrast, all three DAT tests were significantly predicted by a combination of hemispheric asymmetry variables derived from the Linguistic Condition. Table 4 presents these regression results. Hemispheric differences on factors 1, 8, and 11 combined to significantly predict Spatial Relation Test scores ($F(3, 8) = 41.71, p < .00005$). Hemispheric differences on factors 11, 2, and 7 combined to significantly predict Verbal Reasoning Test scores ($F(3, 8) = 36.89, p < .00005$). Hemispheric differences on factors 4 and 11 combined to significantly predict Abstract Reasoning Test scores ($F(2, 9) = 10.00, p < .01$). The multiple correlation coefficients ranged from .83 to .97. It should be noted that hemispheric asymmetry on factor 11, occurring with a relatively late peak latency (416 msec) made significant contributions to the prediction of each of the three cognitive test scores.

DISCUSSION

Both the behavioral and electrophysiological data presented in this paper provide evidence for the existence of functional cerebral specialization in the congenitally deaf population. Task-dependent field advantage asymmetries in accuracy were observed which are consistent with the majority of literature on hearing subjects. Furthermore, two EP components, factors 5 and 8, displayed changes in the relative activity of the two hemispheres as a function of experience in the experimental situation. The hemispheric effects reflected by factor 5 (a very late component) were task dependent. Factor 5 was insensitive to the experimental manipulations in the Linguistic Condition. However, in the Spatial Condition factor 5 maximally discriminated between Same and Different line orientation stimuli during the first half of the condition in the right hemisphere but shifted to a left hemisphere locus of discrimination during the second half of the condition. This shift, which was independent of the field of stimulus presentation, suggests that automatization of processing in this task and population may involve a cooperative transition from right to left hemisphere mechanisms for some late aspect of stimulus discrimination. Factor 8, a very early component, showed a hemispheric

polarity reversal from the first to the second half of the condition, independent of field of stimulation and task. This task-invariant pattern together with the very early post-stimulus latency of factor 8 suggests that the component reflects the activity of a nonspecific processing stage, perhaps sensory or possibly activation in nature. The hemispheric polarity reversal suggests that such nonspecific mechanisms are not necessarily engaged in a bilaterally equivalent manner, but rather, may involve a progressive alteration in relative hemispheric activity as a function of familiarity with tasks and stimuli. Interestingly, these time-dependent shifts in relative hemispheric activity were not paralleled by shifts in visual half-field advantages, indicating that visual half-field data are not necessarily sensitive to some aspects of hemispheric processing during task performance in this population. This conclusion is generally consistent with previous work on hearing subjects (Samar, 1983) using the same experimental paradigm.

Hearing subjects in the previous study showed evidence of task-dependent hemispheric asymmetries on three independent late EP components (between 200 and 600 msec poststimulus) which were stable over the two halves of the conditions. By contrast, no evidence was found in the present deaf sample for time-stable task-dependent asymmetries. It is unclear whether this difference between the two samples is related to differences in underlying cerebral organization, differences in preferred processing strategies or differences on a variety of secondary variables which distinguish hearing and deaf subjects such as language background and educational experience. One commonality in the results from the two studies deserves mention. Factor 2 from the hearing sample in the previous study and factor 5 from the present deaf sample both showed PCA/ANOVA interactions which involved differential hemispheric discrimination of the two classes of stimuli within one of the two experimental tasks. In a general sense, both factors showed task-dependent asymmetries, although the specific pattern of these interactions differed between the two samples. Furthermore, both components had similar latencies of maximum influence on the EPs (factor 2, 552 msec; factor 5, 584 msec). It may be argued on the basis of these similarities that these two components reflect the same general stage of processing in the hearing and deaf groups, a stage which appears to be closely related to more cognitive aspects of stimulus discrimination. If so, this component may be a useful tool in future studies to elaborate similarities and differences between deaf and hearing samples in the cerebral organization of this late stage of stimulus processing.

The results of the regression analyses relating EP asymmetries and visual half-field asymmetries in accuracy during the Spatial and Linguistic tasks underscore the multidimensional character of cerebral asymmetry in the congenitally deaf population. Five EP principal component hemi-

spheric asymmetry variables were required in the Spatial Condition and two in the Linguistic Condition to achieve strong prediction of visual half-field asymmetries. These data are consistent with the notion that visual half-field asymmetries reflect, at best, gross, overall patterns of cerebral specialization. While these results provide further validation of the proposition that visual half-field asymmetries reflect underlying cerebral asymmetries, they simultaneously confirm an element of basic indeterminacy of the precise nature of cerebral specialization given only behavioral asymmetry data (Samar, 1983). In the absence of an adequately elaborated theory of the visual half-field technique, these data suggest that it is not possible to uniquely infer the structure of the underlying multidimensional cerebral asymmetry space from behavioral asymmetry data alone. There are, in principal, an indefinite number of linear equations which could yield the same set of behavioral asymmetry scores observed in the experiment. Taken together with previous research on hearing subjects (Samar, 1983), these results make evident the need for further research directed at an empirically valid and conceptually adequate definition of the relationship between manifest perceptual asymmetries and underlying cerebral asymmetries. This relationship appears to be far more elaborate than current structural (Kimura, 1973) or orientational (Kinsbourne, 1975) models suggest.

This paper also presents evidence for the first time that individual differences in the configuration (in a multidimensional sense) of asymmetrical cerebral function in congenitally deaf adults may be closely related to cognitive performance on an independent set of cognitive tests. It remains unclear why these relationships only appeared when the hemispheric asymmetry predictor variables were derived from the Linguistic Condition. Nevertheless, these results provide a direct impetus for future research on the influence of deafness on cerebral specialization and its possible cognitive consequences.

In general, the present results suggest that cerebral organization in congenitally deaf adults is a complex multidimensional phenomenon which may directly relate to general cognitive performance. Further theoretical advances in this area will depend closely on the use and understanding of techniques and experimental conditions which are capable of resolving the complex structure of the cerebral specialization phenomenon.

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